

META-TRACKER MY

Competitor Sponsorship Intel Station

Field	Detail
Document Version	2.0
System Status	Production Ready
Target Market	Malaysia Technology Media Ecosystem
Deployment Environment	Google AI Studio (React Sandbox Platform)

Section 1: Objective & Strategic Intent

Core Objective

To establish a centralized, zero-cost, privacy-secure competitor intelligence pipeline that extracts unstructured advertising visual assets from the Meta Ads Library and structures them into clean, standardized datasets. The ultimate goal is to map the market penetration, creative velocity, and Share of Voice (SoV) of rival brands across Malaysia's primary technology channels to uncover high-yield business development (BD) sales opportunities and marketing alignment patterns.

Operational Mandate

The tool is designed to eliminate manual data-entry bottlenecks, neutralize translation errors across multilingual ad copies (English, Bahasa Melayu, and Chinese), and provide dynamic timeline visualization.

Section 2: The Tech Stack Ecosystem

The tool leverages a lean, serverless architecture that operates entirely on standard developer free-tiers, incurring zero operational overhead costs:

- Google AI Studio (Playground & App Engine):** Serves as the continuous integration environment and web engine. It converts conversational layout parameters directly into production-grade React application code.
- Gemini 3 Flash Preview / 1.5 Pro (Multimodal Brain):** The underlying foundation model driving the execution. It uses advanced vision capabilities to parse spatial grid files, execute character recognition (OCR), handle semantic mapping, and output clean JSON payloads.
- React Framework Canvas:** The browser frontend layout that maintains the active application state, calculates metrics on the fly, renders visual graphics, and handles user interactions.
- Browser LocalStorage Cache:** Acts as the local database buffer, preserving up to 5,000 text-only records directly inside the client browser to prevent data loss from accidental refreshes.
- Native Inline SVG Graphics Engine:** Renders time-series lines and distribution bars directly using math vectors in Tailwind CSS, eliminating dependencies on heavy external charting frameworks that could crash the playground compiler.

Section 3: System Architecture Blueprint

To reproduce or redeploy this system inside Google AI Studio, your team must apply these absolute, finalized configuration parameters across the two primary workspace input interfaces.

1. The Right-Sidebar System Instructions

Paste this code block directly into the System instructions input field in the right panel to lock down the background parsing filters.

```
# Role
You are a multimodal Ad-Tech Vision & Extraction Engine specializing in the Malaysian technology media landscape.

# Objective
Analyze the uploaded screenshot image of a Meta Ad. Visually execute OCR to extract structured metadata, identify the true brand and phone/service models, normalize the target publisher tags, and return a clean data object.

# Master Publisher Mapping & Language Normalization Matrix
You must classify the "Publisher Tag" field strictly into one of the 10 master parent categories listed below based on the page identity or contextual mentions. Do not use catch-all categories like "Other" or "Others".

Map the incoming variants exactly as follows:
- "Amanz Network", "Amanz" -> Map to: "Amanz"
- "TechNave", "TechNave 中文版", "TechNave BM" -> Map to: "TechNave"
- "Zing Gadget 中文版", "Zing Gadget" -> Map to: "Zing Gadget"
- "ProductNation.co", "Product Nation", "ProductNation" -> Map to: "Product Nation"
- "SoyaCincau.com", "SoyaCincau - BM", "SoyaCincau - 中文版", "Soya Cincau" -> Map to: "Soya Cincau"
- "Hitech Century" -> Map to: "Hitech Century"
- "Pokde.net" -> Map to: "Pokde.net"
- "Adam Lobo TV", "Adam Lobo" -> Map to: "Adam Lobo TV"
- "PriceShop Malaysia", "PriceShop" -> Map to: "PriceShop Malaysia"
- "SAYS" -> Map to: "SAYS"

# Visual OCR Extraction Schema
For every distinct ad card visible in the image, extract and normalize these text fields:
1. Library ID: The unique numerical string (e.g., "1760972465092368").
2. Started Running On: The launch date. Normalize strictly to YYYY-MM-DD.
3. Platform: List visible platforms (e.g., Facebook, Instagram, Messenger).
4. Page Name: The exact name of the account running the ad (e.g., "SoyaCincau - BM" - preserve exact variant for row granularity).
5. Publisher Tag: The normalized master parent string from the Mapping Matrix above.
6. Advertiser / Brand: The actual corporate provider/manufacturer (e.g., ASUS, Samsung, Xiaomi, Maxis).
7. Service / Phone Model: The exact device name or package plan (e.g., ExpertBook Ultra B9406, Galaxy S26 Ultra).
8. Status: Always default to "Active".

# Processing Constraints
- Strict Classification Enforcer: Every analyzed ad must map to one of the 10 verified publisher categories based on matching indicators in the copy or profile page. Do not output "Other".
- Text-Only Memory Protection: Do not store image data inside the output array string. Return pure text properties to guarantee local storage space efficiency.

# Output Format Enforcer
```

Output the parsed results strictly as a valid JSON Array of Objects matching the target fields. Do not add conversational text, introductory notes, or markdown formatting blocks outside the JSON array.

2. The Chatbox Application Prompt

Paste this engineering description into the center chatbox block to generate or refactor the user interface code.

Refactor the application into a highly resilient, two-tab corporate executive workstation built in React with an emerald green and dark slate premium tech aesthetic. The application architecture must strictly implement the following code mechanics:

1. Two-Tab Workspace Layout:

- Tab 1: "Campaign Ingestion & Log" (Data entry, data table, filtration, and storage controls).
- Tab 2: "Executive Insights Hub" (Data visualizations, business development gap tracking, and automated client pitching features).

2. Global Date Control Header:

- Add a horizontal filter bar beneath the main title bar containing HTML5 date inputs for "Start Date" and "End Date", alongside a quick-select dropdown ("All Time", "This Month", "Last 30 Days").
- Altering these parameters must immediately filter both the Tab 1 logs and the Tab 2 dashboards to reflect only campaigns whose 'Started Running On' falls within the range.

3. Data Resilience & Crash-Proof Storage Protocol (Tab 1):

- Anti-Crash Architecture: The master data state array must ONLY store lightweight text fields. Drop heavy Base64 image files from browser memory immediately after OCR completion.
- LocalStorage Persistence: Program a useEffect hook to automatically load stored records from browser local storage on startup and sync state updates automatically.
- Strict Duplicate Prevention: Parse 'Library ID' before appending. If it exists in state, reject ingestion and fire a warning notification toast: "Duplicate Entry Blocked: Ad ID already exists".
- System Storage Health Gauge: Add a text capacity warning card in the UI: "System Capacity: [Total Rows] / 5,000 Records. (Optimized text-only storage structure prevents web-client crashing)."

4. Data Ingestion UI & Table Setup (Tab 1):

- File Input Dropzone: A drag-and-drop file block that captures clipboard paste shortcuts (Ctrl+V/Cmd+V) globally and displays a visual thumbnail preview before rendering a loader spinner.
- Team Protocol Banner: Display a persistent notice text: "Team Protocol Reminder: No weekend shifts required. Use this tool for a quick 5-minute daily pulse or a Monday morning look-back sweep to catch weekend data safely."
- Automated Ingestion Date: Automatically inject a column named "Date Image Uploaded" using the client machine clock formatted as YYYY-MM-DD upon click execution.
- Master Table Columns: Library ID, Started Running On, Date Image Uploaded, Platform, Page Name, Publisher Tag, Advertiser / Brand, Service / Phone Model, and Status.
- Deep Link Generation: Add a clickable external icon link column referencing: `https://www.facebook.com/ads/library/?id=\${libraryId}` for asset verification.
- Editable Cells & Formats: Enable double-click overrides on 'Brand', 'Model', and 'Tag' cells. Provide buttons for "Download CSV Report", "Copy TSV to Clipboard" (optimized for Google Sheets pasting), and a red "Reset Session Data" button.
- Search Bar: Add a textual filter above the log table to filter items instantly.

5. Advanced Executive Marketing & BD Analysis Features (Tab 2):

- Share of Voice (SoV) Grid: Calculate the volume percentage held by each Advertiser/Brand under every master publisher tag and render it as a horizontal, stack-colored distribution bar. Add a secondary widget detailing Platform distribution metrics.

- Campaign Launch Velocity SVG Graph: Render a responsive interactive SVG Line Chart plotting the total number of new campaign launches over daily/weekly coordinate nodes based on the date filter, using emerald gradient strokes and node hover tooltips.

- Historical Monthly Performance Matrix: Aggregate data into chronological months (e.g., "July 2026") and present vertical bar metrics indicating volume shifts.

- Competitor Gap Finder: If a "Service / Phone Model" appears 3+ times globally but has 0 records tied to a specific target publisher, display an alert card: "Market Penetration Opportunity: [Model] is active with competitors but missing from [Publisher]".

- Automated Pitch Generator: Add a dropdown menu to select a target "Brand" and "Publisher", outputting copy-pasteable email copy: "Market Insight: In the last 30 days, your brand held a [X]% Share of Voice across tracked tech channels, but achieved 0% penetration on [Publisher]. Competitors are out-indexing you on this platform by [Y] active campaigns. Let's discuss a targeted activation plan."

Section 4: System Capabilities & App Abilities

1. Zero-Friction Vision OCR Ingestion — The app listens globally to browser pasting hooks. Your team does not have to deal with downloading files, file-picker selectors, or manual text copying. Selecting an image via clipboard commands immediately initializes processing.
2. Multi-Tier Entity Normalization Matrix — The backend includes a hardcoded normalization dictionary. It reads incoming fragmented language indicators (e.g., "SoyaCincau - BM" or "TechNave 中文版") and collapses them down into unified parent tags. This keeps data clean and prevents metric separation across different language variants.
3. Session State Memory Resilience — The application runs on a text-only cache pipeline. By isolating, rendering, and subsequently deleting heavy image arrays from the browser state memory, it avoids QuotaExceededError page crashes. It provides data protection via automated local storage syncs that survive browser crashes or accidental page reloads.
4. Direct Verification Auditing — By converting numerical text keys into direct web references ([https://www.facebook.com/ads/library/?id=\[ID\]](https://www.facebook.com/ads/library/?id=[ID])), the dashboard links every data row back to Meta's live archive servers, allowing for immediate creative validation.

Section 5: Advantages vs. Weaknesses

Core Strategic Advantages

- **Morale Protection (No Weekend Shifts Required):** The framework aligns perfectly with standard weekday corporate parameters. By leveraging Meta's data retention mechanics, your team can capture weekend marketing waves via a strategic Monday morning look-back sweep without working overtime.
- **Flawless Data Ingest Accuracy:** The normalization logic completely removes the "Other/Others" category data leakage problem. The system forces every parsed ad to classify into a clean, tracking-supported profile.
- **Instant Dashboard Execution:** It converts raw visuals into interactive SVG graphs, gap maps, and pre-written business development email templates in under 15 seconds per asset.
- **Zero Software Licensing Cost:** Built, hosted, and maintained inside standard enterprise free-tiers.

Known System Weaknesses & Constraints

- **The Viewport Extraction Trap:** The tool is bound by visual limits. It can only process what is captured in the screenshot snippet. If a competitor scales out 60 active ads simultaneously, the user must execute multiple manual screenshot passes to capture the entire library block.
- **Meta Live-Status Disappearance:** Meta instantly removes active non-political commercial ads the moment the media campaign budget drops to zero. If a competitor runs an agile 24-hour flash sale over the weekend and stops it before Monday morning, the retrospective sweep will miss it.
- **Browser Sandbox Isolation:** Because the system saves data to the client browser's local storage array, the data is isolated to that specific computer. If Team Member A inputs data on Computer 1, Team Member B cannot view it on Computer 2 until the dataset is exported and pushed to the master team Google Sheet.

Section 6: Expected Output Schema

When the data parser executes successfully, it outputs structured data matching these precise validation constraints:

Column Name	Data Type	Constraint / Normalization Rule	Example Value
Library ID	String (Numeric)	Primary Key value extracted from Meta UI layout string.	1760972465092368
Started Running On	Date (YYYY-MM-DD)	The initial launch date parsed from the card header metadata.	2026-07-15
Date Image Uploaded	Date (YYYY-MM-DD)	Auto-generated timestamp from the computer's system clock.	2026-07-18
Platform	String (Array)	List of delivery networks found on the asset card.	Facebook, Instagram
Page Name	String	The exact textual name of the localized media account.	TechNave 中文版
Publisher Tag	String	The normalized parent brand name (limited to the 10 master tags).	TechNave
Advertiser / Brand	String	The corporate entity financing the media buy campaign.	ASUS
Service / Phone Model	String	Normalized romanized product catalog identifier.	ExpertBook Ultra B9406
Status	String (Enum)	Defaults to tracking live delivery states.	Active

Section 7: Training Manual for Teammates

Share this exact step-by-step training sequence with your marketing and data entry team to establish a smooth workflow:

Daily Weekday Pulse Routine (Tuesday to Friday) — Time: 5 Mins

1. Log into your workstation at 9:00 AM. Open the shared tracking app dashboard link.
2. Open a separate browser window and navigate to the Meta Ads Library.

3. Run searches for your assigned competitor profiles using the target list (e.g., TechNave, SoyaCincau, PriceShop).
4. If you spot a fresh ad that was launched yesterday, capture it using your operating system's screenshot tool (Win + Shift + S on Windows or Cmd + Shift + 4 on Mac). Drag the selection box over the entire ad card, making sure the ID Number, Started Running date, and ad body copy text are clearly inside your snippet frame.
5. Click back into the tracking dashboard web page, select the drop zone, and hit Ctrl + V.
6. Click Analyze & Extract Ad Data. Verify that the text drops correctly into the data table log columns.

The Monday Morning Look-Back Sweep Protocol — Time: 15 Mins

Follow this sequence every Monday morning to capture peak weekend advertising spend without working overtime on Saturday or Sunday:

1. Open the Meta Ads Library at 9:15 AM on Monday.
2. Search for the target media publisher accounts. Scroll down the results page chronologically.
3. Check the "Started running on" metadata line at the top of the ad cards.
4. Capture screenshots of every ad that shows a launch date matching Friday, Saturday, or Sunday. Scroll down and keep snipping until you reach an ad that launched before Friday.
5. Paste them into the dashboard page one by one. The application will automatically assign today's date (Monday) to the Date Image Uploaded column, but it will extract the true weekend launch date into the Started Running On column. This keeps our time-series tracking models perfectly accurate.

Data Management & Spreadsheet Syncing

1. At 5:00 PM every Friday afternoon, the designated team operator must click the Copy TSV to Clipboard button on the dashboard interface.
2. Open your team's master shared corporate Google Sheet database.
3. Highlight cell A1 (or the first empty row cell at the bottom of your running log) and press Ctrl + V. The data will automatically split perfectly into your columns.
4. Click the red Reset Session Data button on the web app to clear the local browser state, setting up a fresh canvas for the next week.

Section 8: Roadmap for Future Technical Improvements

As the tracking volume scales over the coming months, implement these architectural upgrades to transition from a local sandbox app into a fully automated enterprise database:

[React App Canvas] —> Post Event Request —> [Google Apps Script Webhook API] —> [Master Google Sheet Repository]

- **Google Sheets Webhook Ingestion Engine:** Replace the manual Copy-to-Clipboard step by wiring up a direct backend database connection. By deploying a small, free Google Apps Script Webhook API inside your master Google Sheet, you can modify the React app's action button to automatically send every processed text row directly to the sheet over the internet in real-time. This allows the dataset to update automatically across different team members' computers.

- **True Database Persistence:** When the volume eventually crosses 10,000 campaigns, scale the text data out of Google Sheets and pipe it into a free cloud database like Supabase or Vercel Postgres to ensure your analytics dashboards load instantly.

